NAT

Algorithm Empirical Results

Walk-Forward Out-of-Sample Paper Trading

18 Algorithms, 3 Symbols, 13 OOS Dates

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Contents

1 Test Conditions

All algorithms were evaluated under identical walk-forward out-of-sample (OOS) conditions:

- **Data interval:** 2026-04-19 to 2026-05-23 (19 dates, ∼35 calendar days)
- **OOS window:** 13 dates (2026-05-07 to 2026-05-23)
- **Tick resolution:** 100ms (∼108k rows/day/symbol)
- **Bar resolution:** 5 minutes (aggregated from ticks)
- **Horizon:** 100 minutes (20 bars)
- **Fee model:** 1.61 bps round-trip (Binance VIP9 taker)
- **Symbols:** BTC, ETH, SOL (Hyperliquid perpetual futures)
- **Training window:** 3-day rolling z-score calibration
- **Entry thresholds:** P20/P80 percentile of z-scored feature
- **Exit:** Fixed 100-minute hold, mark-to-market close

Walk-forward protocol: For each OOS date d, the z-score parameters (mean and standard deviation) are calibrated on dates $\{d-3, d-2, d-1\}$. The signal is applied to date d using these frozen parameters. No information from date d or later enters the calibration.

2 Algorithm Ranking

Table 1 presents all 18 algorithms ordered by total OOS PnL across all three symbols.

Table 1: Algorithm ranking by total OOS PnL (bps). Sharpe ratios are annualized ($\times\sqrt{252}$).

#	Algorithm	BTC (bps)	BTC SR	ETH (bps)	ETH SR	SOL (bps)	SOL SR	Total
1	**jump_detector**	+1,722	+1.6	+10,861	+6.2	+10,616	+6.2	**+23,199**
2	**3f_liquidity**	+5,302	+9.2	+7,162	+7.8	+3,564	+3.2	**+16,028**
3	**funding_rev.**	+429	+0.4	+10,265	+6.1	+3,766	+1.7	**+14,459**
4	**optimal_entry**	+1,504	+1.1	+9,877	+5.2	+2,298	+1.0	**+13,679**
5	surprise_signal	−4,563	−8.3	+2,878	+3.1	+5,190	+6.7	+3,505
6	oi_divergence	−1,648	−5.3	−2,596	−5.7	+2,523	+2.1	−1,721
7	regime_gated	−1,312	−2.4	−402	−0.4	−34	−0.0	−1,748
8	entropy_mom.	−1,412	−6.4	−98	−0.2	−1,090	−2.7	−2,600
9	propagator	−641	−2.4	−771	−1.2	−2,706	−3.9	−4,118
10	hawkes_intens.	+409	+0.8	−1,800	−2.6	−4,052	−4.3	−5,443
11	trade_through	−2,437	−5.1	−3,444	−4.2	+142	+0.2	−5,739
12	multi_level_imb	−2,981	−5.4	−1,130	−1.5	−1,699	−1.9	−5,810
13	weighted_ofi	−2,667	−4.8	−430	−0.6	−3,086	−3.6	−6,183
14	switching_ou	−1,341	−3.5	+462	+0.7	−5,351	−6.0	−6,230
15	vpin_regime	−2,561	−4.7	−1,405	−1.7	−3,365	−3.5	−7,331
16	kalman_imb.	−1,328	−2.4	−14	−0.0	−6,175	−7.2	−7,517
17	bipower_jump	−9,161	−14.0	−9,744	−9.7	−13,174	−9.8	−32,079
18	spread_decomp	−10,861	−10.7	−13,249	−14.8	−10,400	−8.4	−34,510

Tier classification:

- **Tier 1** (ranks 1–4): Net positive across all three symbols. Deployable.
- **Tier 2** (rank 5): Symbol-specific alpha (ETH, SOL only).
- **Tier 3** (ranks 6–18): Net negative in aggregate. No edge after costs.

3 Tier 1 Algorithms

3.1 Jump Detector — Post-Jump Mean Reversion

Core idea: Detect price jumps using the Lee–Mykland (2008) nonparametric test, then enter a mean-reversion position after the jump dissipates.

Mathematical Framework

Bipower variation provides a jump-robust volatility estimator:

$$\hat{\sigma}_{\mathrm{BV}} = \sqrt{\frac{\pi}{2} \cdot \frac{1}{n-1} \sum_{i=2}^{n} |r_i| \cdot |r_{i-1}|} \tag{1}$$

where $r_i = \log(p_i/p_{i-1})$ are log-returns. The constant $\mu_1 = \sqrt{2/\pi}$ arises from $\mathbb{E}[|Z|]$ for $Z \sim \mathcal{N}(0,1)$.

Lee–Mykland $\mathcal{L}$-statistic:

$$\mathcal{L}_t = \frac{r_t}{\hat{\sigma}_{\mathrm{BV},t}} \tag{2}$$

Under the null (no jump), $|\mathcal{L}_t|$ follows the distribution of the maximum of i.i.d. standard normals. A jump is declared when:

$$\frac{|\mathcal{L}_t| - C_n}{S_n} > \beta_{\mathrm{threshold}} \tag{3}$$

where $C_n = \sqrt{2\log n} - \frac{\log \pi + \log\log n}{2\sqrt{2\log n}}$ and $S_n = \frac{1}{\sqrt{2\log n}}$ are the Gumbel centering and scaling constants.

Signal: After a jump is detected, the reversion signal activates:

$$s_t = -\mathrm{sign}(\mathcal{L}_t) \cdot \mathbb{1}\{\text{jump detected at } t\} \tag{4}$$

The primary feature `alg_post_jump_reversion` is the exponentially decayed reversion signal (half-life = 50 bars).

Table 2: Jump detector OOS performance.

Symbol	Trades	Net bps/trade	Sharpe (ann.)	Total PnL	Win Rate	Max Daily Loss
BTC	1,678	+1.03	+1.6	+1,722	54%	−1,981
ETH	1,678	+6.47	+6.2	+10,861	62%	−3,412
SOL	1,678	+6.33	+6.2	+10,616	69%	−3,311

Implementation: `scripts/algorithms/jump_detector.py`

3.2 3f Liquidity Signal — Composite Microstructure

Core idea: Equal-weight z-score composite of three liquidity features, capturing the overall microstructure state.

Signal Construction

Three base features are computed at 5-minute bar resolution:

$$f_1 = \text{spread\_bps} \quad \text{(bid-ask spread in basis points)} \tag{5}$$

$$f_2 = \text{depth\_imbalance} \quad \text{(bid vs. ask depth ratio)} \tag{6}$$

$$f_3 = \text{vwap\_deviation} \quad \text{(price distance from VWAP)} \tag{7}$$

Each feature is z-scored over a 3-day rolling window:

$$z_i(t) = \frac{f_i(t) - \hat{\mu}_i}{\hat{\sigma}_i}, \quad \hat{\mu}_i, \hat{\sigma}_i \text{ from } [t - 3\text{d}, t) \tag{8}$$

The composite signal is the equal-weight average:

$$S(t) = \frac{1}{3} \sum_{i=1}^{3} z_i(t) \tag{9}$$

Entry: Long when $S(t) < P_{20}$, short when $S(t) > P_{80}$ (percentiles from training window).

Table 3: 3f liquidity signal OOS performance.

Symbol	Trades	Net bps/trade	Sharpe (ann.)	Total PnL	Win Rate
BTC	950	+5.58	+9.2	+5,302	62%
ETH	915	+7.83	+7.8	+7,162	62%
SOL	954	+3.74	+3.2	+3,564	62%

Implementation: `scripts/alpha/paper_trader.py`

3.3 Funding Reversion — Crypto-Native Mean Reversion

Core idea: When the perpetual funding rate deviates from its mean, it tends to revert. Extreme funding = crowded positioning = reversion opportunity.

Signal Construction

The funding rate is z-scored over a rolling window, then passed through a saturating function:

$$S(t) = -\text{sign}(z_t) \cdot \min\left(\frac{|z_t|}{z_{\text{entry}}}, 3\right) \Big/ 3 \tag{10}$$

where $z_t = (r_t^{\text{fund}} - \hat{\mu})/\hat{\sigma}$ is the z-scored funding rate and z_{entry} is the entry threshold.

The saturation at ± 1 prevents excessive signal magnitude during extreme funding events.

Table 4: Funding reversion OOS performance.

Symbol	Trades	Net bps/trade	Sharpe (ann.)	Total PnL	Win Rate	Max Daily Loss
BTC	1,678	+0.26	+0.4	+429	38%	−2,327
ETH	1,678	+6.12	+6.1	+10,265	54%	−2,629
SOL	1,678	+2.24	+1.7	+3,766	54%	−3,786

Implementation: `scripts/algorithms/funding_reversion.py`

3.4 Optimal Entry — SPRT on Kalman Innovations

Core idea: Model L1 order book imbalance as an Ornstein–Uhlenbeck process, filter with a Kalman filter, then apply a Sequential Probability Ratio Test (SPRT) on the innovations to determine entry timing.

Kalman–OU Filter

The imbalance x_t follows an OU process:

$$dx_t = \kappa(\theta - x_t)\,dt + \sigma\,dW_t \tag{11}$$

Discretized with step Δt:

$$\hat{x}_{t|t-1} = \theta + e^{-\kappa\Delta t}(\hat{x}_{t-1} - \theta) \tag{12}$$

$$P_{t|t-1} = e^{-2\kappa\Delta t}P_{t-1} + \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta t}) \tag{13}$$

The innovation (prediction error) is:

$$\nu_t = x_t - \hat{x}_{t|t-1} \tag{14}$$

SPRT

The log-likelihood ratio accumulates evidence for a directional drift μ in the innovations:

$$\Lambda_t = \Lambda_{t-1} + \frac{\mu}{\sigma_\nu^2}\,\nu_t - \frac{\mu^2}{2\sigma_\nu^2} \tag{15}$$

Decision boundaries (Wald, 1945):

$$A = \log\frac{1-\beta}{\alpha}, \quad B = \log\frac{\beta}{1-\alpha} \tag{16}$$

Signal: $s_t = +1$ when $\Lambda_t \geq A$ (long), $s_t = -1$ when $\Lambda_t \leq B$ (short), $s_t = 0$ otherwise.

Table 5: Optimal entry OOS performance.

Symbol	Trades	Net bps/trade	Sharpe (ann.)	Total PnL	Win Rate	Max Daily Loss
BTC	1,678	+0.90	+1.1	+1,504	46%	−2,327
ETH	1,678	+5.89	+5.2	+9,877	62%	−3,645
SOL	1,678	+1.37	+1.0	+2,298	54%	−4,762

Implementation: `scripts/algorithms/optimal_entry.py`
Known issue: `run_batch()` hardcodes `sigma_process=0.01` at line 240 instead of using `self._sigma_process`. This affects batch–live equivalence.

4 Tier 2: Symbol-Specific Alpha

4.1 Surprise Signal — Entropy Regime Transitions

Core idea: Track the rate of change of order book entropy. A sudden drop (ordering event) signals informed participants entering; a sudden rise (disordering) signals noise.

Signal Construction

Given entropy time series H_t (e.g., book shape entropy), compute the rate of change:

$$\mathrm{ROC}_t = H_t - H_{t-\tau} \tag{17}$$

Z-score the ROC over a rolling window:

$$z_t^{\mathrm{ROC}} = \frac{\mathrm{ROC}_t - \hat{\mu}_{\mathrm{ROC}}}{\hat{\sigma}_{\mathrm{ROC}}} \tag{18}$$

Map to a regime transition probability via logistic sigmoid:

$$P_{\text{transition}}(t) = \frac{1}{1 + \exp\big(-(|z_t^{\text{ROC}}| - z_0)\big)} \tag{19}$$

Signal polarity: Negative surprise (ordering) → long. Positive surprise (disordering) → short.

Table 6: Surprise signal OOS performance.

Symbol	Trades	Net bps/trade	Sharpe (ann.)	Total PnL	Win Rate	Max Daily Loss
BTC	954	−4.78	−8.3	−4,563	15%	−1,316
ETH	1,010	+2.85	+3.1	+2,878	54%	−2,493
SOL	981	+5.29	+6.7	+5,190	46%	−518

Implementation: `scripts/algorithms/surprise_signal.py`

5 Statistical Certainty

5.1 Sharpe Ratio Confidence Intervals

The annualized Sharpe ratio $\widehat{SR}_{\text{ann}}$ is estimated from n daily observations. Following Lo (2002), its standard error under i.i.d. returns is:

$$\text{SE}(\widehat{SR}_{\text{ann}}) = \sqrt{\frac{1 + \frac{1}{2}\widehat{SR}_{\text{daily}}^2}{n}} \cdot \sqrt{252} \tag{20}$$

where $\widehat{SR}_{\text{daily}} = \widehat{SR}_{\text{ann}}/\sqrt{252}$.

The 95% confidence interval is:

$$\widehat{SR}_{\text{ann}} \pm 1.96 \cdot \text{SE}(\widehat{SR}_{\text{ann}}) \tag{21}$$

Table 7: Sharpe ratio confidence intervals ($n = 13$ daily observations, 95% CI).

Algorithm	Symbol	$\widehat{SR}_{\text{ann}}$	SE	95% CI Lower	95% CI Upper
jump_detector	BTC	+1.62	4.41	−7.03	+10.27
jump_detector	ETH	+6.23	4.57	−2.73	+15.19
jump_detector	SOL	+6.22	4.57	−2.73	+15.17
3f_liquidity	BTC	+9.22	4.76	−0.11	+18.55
3f_liquidity	ETH	+7.83	4.66	−1.31	+16.97
3f_liquidity	SOL	+3.22	4.44	−5.50	+11.94
funding_reversion	BTC	+0.38	4.40	−8.25	+9.01
funding_reversion	ETH	+6.05	4.56	−2.89	+14.99
funding_reversion	SOL	+1.69	4.41	−6.96	+10.34
optimal_entry	BTC	+1.12	4.40	−7.52	+9.76
optimal_entry	ETH	+5.23	4.52	−3.63	+14.09
optimal_entry	SOL	+0.98	4.40	−7.66	+9.62
surprise_signal	BTC	−8.29	4.72	−17.49	+0.91
surprise_signal	ETH	+3.12	4.44	−5.59	+11.83
surprise_signal	SOL	+6.70	4.60	−2.31	+15.71

Key observation: With only $n = 13$ daily observations, *no individual algorithm–symbol pair achieves a 95% CI that excludes zero.* The best case is 3f_liquidity on BTC ($\widehat{SR} = 9.2$, CI lower bound $= -0.11$), which narrowly misses. This is a fundamental sample-size limitation, not evidence against the signals.

5.2 Hypothesis Testing on Daily PnL

For each algorithm–symbol pair, we test:

$$H_0 : \mu_{\text{daily}} = 0 \quad \text{vs.} \quad H_1 : \mu_{\text{daily}} \neq 0 \tag{22}$$

The test statistic is:

$$t = \frac{\bar{r}}{s/\sqrt{n}} = \widehat{SR}_{\text{daily}} \cdot \sqrt{n} \tag{23}$$

where $\bar{r}$ is the mean daily PnL, s is the sample standard deviation, and $n = 13$.

Table 8: t-test on daily PnL ($n = 13$, df $= 12$). Critical values: $t_{0.05} = 2.179$, $t_{0.01} = 3.055$.

Algorithm	Symbol	$\widehat{SR}_{\text{daily}}$	t-statistic	p-value	Significant?
3f_liquidity	BTC	+0.581	+2.094	0.058	marginal
3f_liquidity	ETH	+0.493	+1.778	0.101	no
3f_liquidity	SOL	+0.203	+0.731	0.479	no
jump_detector	BTC	+0.102	+0.368	0.719	no
jump_detector	ETH	+0.393	+1.415	0.182	no
jump_detector	SOL	+0.392	+1.413	0.183	no
funding_rev.	ETH	+0.381	+1.374	0.195	no
optimal_entry	ETH	+0.330	+1.188	0.258	no
surprise_signal	SOL	+0.422	+1.522	0.154	no

Result: No algorithm achieves $p < 0.05$ with $n = 13$ days. The closest is 3f_liquidity on BTC ($t = 2.09$, $p = 0.058$), which is marginally significant.

5.3 Multiple Testing Correction

With 18 algorithms $\times$ 3 symbols = 54 simultaneous tests, we must correct for multiple comparisons.

Bonferroni Correction

The family-wise error rate is controlled at $\alpha = 0.05$ by adjusting the per-test significance level:

$$\alpha^* = \frac{\alpha}{m} = \frac{0.05}{54} \approx 0.000926 \tag{24}$$

Result: No algorithm–symbol pair achieves $p < \alpha^*$ with $n = 13$. This is expected — Bonferroni with 54 tests and 13 observations is extremely conservative.

Benjamini–Hochberg FDR

The Benjamini–Hochberg procedure controls the false discovery rate at $q = 0.05$. Order the $m = 54$ p-values as $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$ and find the largest k such that:

$$p_{(k)} \leq \frac{k}{m} \cdot q \tag{25}$$

All hypotheses $\{1, 2, \ldots, k\}$ are rejected.

Result: With $n = 13$, no discoveries survive BH correction at $q = 0.05$.

5.4 Required Sample Size for Significance

Given the sample-size bottleneck, we compute the minimum number of OOS days n^* required for a two-sided t-test at significance α and power $1 - \beta$:

$$n^* = \left\lceil \left(\frac{z_{1-\alpha/2}}{\widehat{SR}_{\text{daily}}} \right)^2 \right\rceil \tag{26}$$

Table 9: Required OOS days for statistical significance (two-sided, power not considered).

Annualized Sharpe	Days for $p < 0.05$	Days for $p < 0.01$
$SR = 1$	969	1,673
$SR = 3$	108	186
$SR = 5$	39	67
$SR = 7$	20	35
$SR = 9$	12	21

Interpretation: An algorithm with $SR_{\text{ann}} = 6$ (our typical Tier 1 value) requires ~27 OOS days for $p < 0.05$. Our current 13 days is approximately half the required sample. This provides a concrete data collection target: **accumulate ≥30 OOS days before drawing definitive conclusions**.

5.5 Signal Independence

To verify that the winning algorithms provide genuine diversification (not redundant trades), we compute pairwise Spearman rank correlation of trade direction vectors.

Table 10: Spearman direction correlation (BTC, $n = 2{,}022$ OOS bars).

	3f_liq.	jump_det.	fund_rev.	surprise
3f_liquidity	1.000	0.192	0.083	0.338
jump_detector	0.192	1.000	−0.005	0.001
funding_rev.	0.083	−0.005	1.000	0.093
surprise	0.338	0.001	0.093	1.000

Key finding: Jump detector ↔ funding reversion correlation is $\rho = -0.005$ (essentially zero). These are genuinely independent signals. The highest correlation is 3f_liquidity ↔ surprise at $\rho = 0.338$, indicating moderate overlap.

6 Portfolio Implications

6.1 Complementarity

The Tier 1 algorithms show natural complementarity across symbols:

Table 11: Best algorithm per symbol (by annualized Sharpe ratio).

Symbol	Best Algorithm	Sharpe (ann.)
BTC	3f_liquidity	+9.2
ETH	3f_liquidity	+7.8
SOL	surprise_signal	+6.7

A simple blend of 3f_liquidity (all symbols) + jump_detector (ETH/SOL) + surprise_signal (SOL) would exploit each algorithm where it is strongest.

6.2 Diversification Benefit

For two uncorrelated signals with individual Sharpe ratios SR_1 and SR_2, the portfolio Sharpe ratio under equal risk parity is:

$$SR_{\text{portfolio}} = \sqrt{SR_1^2 + SR_2^2} \tag{27}$$

For jump_detector ETH ($SR = 6.2$) and funding_reversion ETH ($SR = 6.1$) with $\rho \approx 0$:

$$SR_{\text{blend}} = \sqrt{6.2^2 + 6.1^2} \approx 8.7 \tag{28}$$

This exceeds either individual signal, confirming the diversification benefit of combining independent microstructure signals.

7 Limitations & Caveats

1. **Small sample size.** With $n = 13$ OOS days, no individual algorithm–symbol pair achieves statistical significance at $\alpha = 0.05$. The point estimates (Sharpe ratios, PnL) are suggestive but not conclusive. Minimum 30 OOS days recommended.

2. **Regime dependency.** The test window (May 7–23, 2026) covers a specific market regime. Algorithms may behave differently in trending, ranging, or high-volatility regimes. Multi-regime OOS testing is required.

3. **Z-score calibration look-ahead.** While the walk-forward protocol prevents direct look-ahead, the choice of 3-day training window and P20/P80 thresholds was itself informed by preliminary analysis on this dataset. A fully independent validation would require a held-out date range.

4. **Fee model.** The 1.61 bps round-trip assumes Binance VIP9 taker rates. At higher fee levels (e.g., 3.5 bps for VIP0, 7 bps for Hyperliquid), several Tier 1 algorithms would become marginal.

5. **Execution assumptions.** The paper trader assumes instant fills at mid-price + half-spread. Real execution faces slippage, queue priority, and latency. Market impact is not modeled.

6. **Survivorship in algorithm selection.** The 18 algorithms were selected from a larger set of ideas based on prior feature analysis. This selection process introduces a form of survivorship bias that is not captured by the multiple testing correction.

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